

corrigé zakito

Énoncé

1

21 atouts, 14 cartes par couleur, 1 excuse.

1. $\binom{14}{2}^2 \binom{21}{3}$.
2. $\binom{21}{3} \binom{14}{2} \binom{43}{2}$.
3. $\binom{14}{7} + 63 \binom{14}{5} + \binom{14}{6}$.
4. $\binom{21}{5} \left(\binom{43}{2} + 14 \cdot 43 \right) + \binom{21}{6} (43 + 14) + \binom{21}{7}$.

2

RIQUIQUI : $(R, I, Q, U) \mapsto (1, 3, 2, 2) \Rightarrow \frac{8!}{3!2!2!} = 1680$.

ABRACADABRA : $(A, B, R, C, D) \mapsto (5, 2, 2, 1, 1) \Rightarrow \frac{11!}{5!2!2!} = 83160$.

3

1. $\binom{np}{n}$, 2. p^n , 3. $\binom{p}{2}^n$.
4. $a_{i, \sigma(i)} = 1 \quad (\sigma \in \mathfrak{S}_n) \Rightarrow n!$.

4

1. $2 \times n! \times n! = 2(n!)^2$.
2. $\frac{2(n!)^2}{2n} = (n-1)n! \quad (n \geq 1)$.

Rotations identifiées, sens de rotation conservé.

5

1. $y = n - x, \quad 1 \leq x \leq n - 1 \Rightarrow n - 1$.
2. $\sum_{y=1}^n (y-1) = \binom{n}{2}$.
3. $\sum_{x=1}^n (2n-x) = \frac{n(3n-1)}{2}$.
4. $n + 2(n-1) = 3n - 2$.

6

$A_i = \{\text{mots de longueur 9 avec le motif commençant en position } i\}$.

1. $|A_i| = 26^4, \quad 1 \leq i \leq 5, \quad A_i \cap A_j = \emptyset \quad (i \neq j) \Rightarrow 5 \cdot 26^4$.
2. $|A_i| = 26^5, \quad 1 \leq i \leq 6$.

$A_i \cap A_j \neq \emptyset \iff j - i \geq 4 \quad (i < j), \quad \#\{(i, j) : j - i \geq 4\} = 3$.

$$|A_i \cap A_j| = 26 \Rightarrow 6 \cdot 26^5 - 3 \cdot 26.$$

$$3. \quad A_i \cap A_j \neq \emptyset \iff j - i = 3 \text{ ou } j - i \geq 4.$$

$$j - i = 3 : \quad 3 \text{ couples}, \quad |A_i \cap A_j| = 26^2; \quad j - i \geq 4 : \quad 3 \text{ couples}, \quad |A_i \cap A_j| = 26.$$

$$\text{Aucune intersection triple} \Rightarrow 6 \cdot 26^5 - 3 \cdot 26^2 - 3 \cdot 26.$$

7

$$\frac{n(n-3)}{2} = n, \quad n \geq 3 \iff n = 5.$$

8

$$N = \prod_{i=1}^r p_i^{\alpha_i}, \quad d \mid N \iff d = \prod_{i=1}^r p_i^{\beta_i}, \quad 0 \leq \beta_i \leq \alpha_i.$$

$$\tau(N) = \prod_{i=1}^r (\alpha_i + 1).$$

9

$$\underbrace{\binom{m}{p}}_{\text{filles}} \underbrace{\binom{n}{p}}_{\text{garçons}} \underbrace{(2p)!}_{\text{classement}}.$$

10

$$f^2 = \text{id} \Rightarrow \{1, \dots, 2n+1\} = \text{Fix}(f) \sqcup \bigsqcup_{j=1}^r \{x_j, f(x_j)\}, \quad f(x_j) \neq x_j.$$

$$2n+1 = |\text{Fix}(f)| + 2r \Rightarrow |\text{Fix}(f)| \text{ impair} \Rightarrow |\text{Fix}(f)| \geq 1.$$

11

$$1. \quad n \geq 1 : \quad 2^n - 2; \quad n = 0 : \quad 0.$$

$$2. \quad n \geq 1 : \quad \underbrace{\binom{n+1}{2}}_{\text{fibre double}} \underbrace{n!}_{\text{images des } n \text{ fibres}}; \quad n = 0 : \quad 0.$$

12

$$1. \quad p 2^{n-p}, \quad 2. \quad (2^p - 1) 2^{n-p} = 2^n - 2^{n-p}.$$

13

$$1. \quad f \longleftrightarrow \{f(1), \dots, f(p)\} \Rightarrow \binom{n}{p}.$$

$$2. \quad 1 \leq f(1) \leq \dots \leq f(p) \leq n$$

$$\iff 1 \leq f(1) < f(2) + 1 < \dots < f(p) + p - 1 \leq n + p - 1 \Rightarrow \binom{n+p-1}{p}.$$

14

$$1. \quad R \subset E^2 \Rightarrow 2^{n^2}.$$

$$2. \quad \{(x, x) : x \in E\} \subset R \Rightarrow 2^{n^2-n}.$$

$$3. \quad x \neq y : (x, y) \in R \iff (y, x) \in R \Rightarrow 2^{\binom{n}{2}}.$$

15

1. $(\mathbf{1}_A(x), \mathbf{1}_B(x)) \in \{(1, 0), (0, 1)\} \Rightarrow 2^n.$
2. $(\mathbf{1}_A(x), \mathbf{1}_B(x)) \in \{(0, 0), (1, 0), (0, 1)\} \Rightarrow 3^n.$
3. $(\mathbf{1}_A(x), \mathbf{1}_B(x)) \in \{(1, 1), (1, 0), (0, 1)\} \Rightarrow 3^n.$

16

$$A_1 \subseteq \dots \subseteq A_p, \quad r(x) = \min\{i : x \in A_i\}, \quad r(x) = p+1 \text{ si } x \notin A_p.$$

$$r : E \rightarrow \{1, \dots, p+1\}, \quad A_i = \{x : r(x) \leq i\}.$$

$$(p+1)^{|E|}.$$

17

1. $4^n.$
 2. $u = x + y, \quad v = x - y, \quad (\Delta u, \Delta v) \in \{(-1, -1), (-1, 1), (1, -1), (1, 1)\}.$
- $\Delta u_{\text{total}} = \Delta v_{\text{total}} = 0 \iff \text{chaque coordonnée comporte } n \text{ pas } +1 \text{ et } n \text{ pas } -1.$
- $$\binom{2n}{n}^2.$$

18

$$N_{p,n} = \#\{f : \{1, \dots, p\} \rightarrow \{1, \dots, n\}\}.$$

$$N_{p,n} = \sum_{J \subseteq \{1, \dots, n\}} (-1)^{|J|} (n - |J|)^p = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} k^p.$$

1. $N_{n,n} = n!, \quad 2. \quad N_{p,n} = \begin{cases} 0 & 0 \leq p < n, \\ n! & p = n. \end{cases}$

19

$$A_i = \{\sigma : \sigma(i) = i\}, \quad \left| \bigcap_{i \in J} A_i \right| = (n - |J|)!.$$

$$d_n = \sum_{J \subseteq \{1, \dots, n\}} (-1)^{|J|} (n - |J|)! = \sum_{k=0}^n (-1)^k \binom{n}{k} (n - k)! = n! \sum_{k=0}^n \frac{(-1)^k}{k!}.$$

20

$$[X^{2n}](1+X)^n(1+X)^{2n} = \sum_{k=0}^n \binom{n}{k} \binom{2n}{2n-k} = \binom{3n}{2n} = \binom{3n}{n}.$$

21

1. $\max A \in \{p+1, \dots, n+p+1\}.$
 2. $\#\{A : |A| = p+1, \max A = m\} = \binom{m-1}{p}.$
- $$\sum_{m=p+1}^{n+p+1} \binom{m-1}{p} = \sum_{k=0}^n \binom{p+k}{p} = \binom{n+p+1}{p+1}.$$

22

1

$$\#\{(A, B) : A \subseteq B \subseteq E, |A| = k, |B| = p\} = \binom{n}{p} \binom{p}{k} = \binom{n}{k} \binom{n-k}{p-k}.$$

2

$$\binom{n}{k} \binom{k}{j} = \binom{n}{j} \binom{n-j}{k-j}.$$

$$S(Tx)_n = \sum_{j=0}^n \binom{n}{j} x_j \sum_{k=j}^n (-1)^{k-j} \binom{n-j}{k-j} = x_n.$$

$$T(Sx)_n = \sum_{j=0}^n \binom{n}{j} x_j \sum_{k=j}^n (-1)^{n-k} \binom{n-j}{k-j} = x_n.$$

$$S \circ T = T \circ S = \text{id}_{\mathbb{R}^{\mathbb{N}}}.$$

23

$$\#\{(x, A) \in E \times D : x \in A\} = \sum_{x \in E} n = n^2 p = \sum_{A \in D} p = p|D|.$$

$$p \geq 1 \Rightarrow |D| = n^2.$$

24

1(a)

$$s_1 = 1, \quad s_2 = 2, \quad s_3 = 4.$$

Première entrée j : $j = n+1 \Rightarrow 1$ liste; $1 \leq j \leq n \Rightarrow s_{n+1-j}$ listes.

$$s_{n+1} = 1 + \sum_{k=1}^n s_k.$$

Le s_0 imprimé n'est pas défini ; cette formule lui correspond avec $s_0 = 0$.

1(b)

$$n \geq 2 : \quad s_{n+1} - s_n = s_n, \quad s_2 = 2s_1 \Rightarrow s_n = 2^{n-1}.$$

1(c)

$$\underbrace{1 \ 1 \ \dots \ 1}_{n \text{ unités}}, \quad n-1 \text{ interstices}, \quad \text{séparer ou réunir à chaque interstice} \Rightarrow 2^{n-1}.$$

2(a)

$$s_{1,p} = \mathbf{1}_{p=1}, \quad s_{2,p} = \mathbf{1}_{p=1} + \mathbf{1}_{p=2}, \quad s_{n,1} = 1, \quad s_{n,2} = n-1.$$

2(b)

$$a_{p+1} = 1 \Rightarrow (a_1, \dots, a_p) \text{ de somme } n;$$

$$a_{p+1} \geq 2 \Rightarrow (a_1, \dots, a_p, a_{p+1}-1) \text{ de somme } n.$$

$$s_{n+1,p+1} = s_{n,p} + s_{n,p+1}.$$

2(c)

$$\binom{n-1}{p-1} + \binom{n-1}{p} = \binom{n}{p}, \quad s_{n,1} = 1, \quad s_{1,p} = \mathbf{1}_{p=1}.$$

Récurrance sur $n \Rightarrow s_{n,p} = \binom{n-1}{p-1}$.

2(d)

p parts non vides $\longleftrightarrow p-1$ séparateurs parmi $n-1$ interstices $\Rightarrow \binom{n-1}{p-1}$.

25

Entiers distincts.

$$1. \quad \{1, \dots, 2n\} = \bigsqcup_{j=1}^n \{2j-1, 2j\}.$$

$n+1$ éléments, n paires $\Rightarrow \exists j, \quad 2j-1, 2j$ choisis.

$$2. \quad m = 2^{v_2(m)} u(m), \quad u(m) \in \{1, 3, \dots, 2n-1\}.$$

$$\exists a \neq b, \quad u(a) = u(b) \Rightarrow a = 2^r u, \quad b = 2^s u \Rightarrow a \mid b \text{ ou } b \mid a.$$

26

v fixé, 5 voisins, 2 types de lien $\Rightarrow \exists a, b, c$ liés à v par le même type T .

$\exists$ lien de type T parmi $a, b, c \Rightarrow$ triangle de type T avec v .

Sinon, a, b, c forment un triangle de l'autre type.

27

1

$$S_0 = 0, \quad S_j = \sum_{k=1}^j x_k.$$

$n+1$ sommes, n résidus $\Rightarrow \exists 0 \leq i < j \leq n, \quad S_j \equiv S_i \pmod{n}$.

$$I = \{i+1, \dots, j\} \neq \emptyset, \quad \sum_{k \in I} x_k \in n\mathbb{Z}.$$

2

$$t_j = S_j - \lfloor S_j \rfloor \in [0, 1), \quad t_{\sigma(0)} \leq \dots \leq t_{\sigma(n)}.$$

$$\sum_{r=0}^{n-1} (t_{\sigma(r+1)} - t_{\sigma(r)}) + (1 + t_{\sigma(0)} - t_{\sigma(n)}) = 1.$$

$$\min \text{ des } n+1 \text{ écarts circulaires} \leq \frac{1}{n+1}.$$

$$\exists 0 \leq i < j \leq n, \quad \exists N \in \mathbb{Z}, \quad |S_j - S_i - N| \leq \frac{1}{n+1}.$$

$$I = \{i+1, \dots, j\}.$$

28

1

$$[0, 1) = \bigsqcup_{r=0}^{n-1} [r/n, (r+1)/n).$$

$$\delta_0, \dots, \delta_n \Rightarrow \exists 0 \leq i < j \leq n, \quad |\delta_j - \delta_i| < 1/n.$$

$$q = j - i, \quad p = \lfloor jx \rfloor - \lfloor ix \rfloor \Rightarrow 1 \leq q \leq n, \quad \left| x - \frac{p}{q} \right| < \frac{1}{nq}.$$

2(a)

$$Q \geq 1, \quad x \notin \mathbb{Q} \Rightarrow \varepsilon_Q = \min_{1 \leq q \leq Q} \text{dist}(qx, \mathbb{Z}) > 0.$$

$$n > 1/\varepsilon_Q \Rightarrow |q_n x - p_n| < 1/n < \varepsilon_Q \Rightarrow q_n > Q.$$

$$q_n \leq n \Rightarrow \left| x - \frac{p_n}{q_n} \right| < \frac{1}{nq_n} \leq \frac{1}{q_n^2}.$$

$$q_n \rightarrow +\infty \Rightarrow \text{une infinité de couples.}$$

2(b)

$$|p_n| \geq |x|q_n - |p_n - q_n x| > |x|q_n - 1/q_n \rightarrow +\infty.$$

29**1**

$$m = [kp] = [\ell q] \Rightarrow m < kp < m+1, \quad m < \ell q < m+1.$$

$$m \left(\frac{1}{p} + \frac{1}{q} \right) < k + \ell < (m+1) \left(\frac{1}{p} + \frac{1}{q} \right) \Rightarrow m < k + \ell < m+1, \quad \perp.$$

2

$$[kp] \leq n \iff kp < n+1 \iff k < \frac{n+1}{p} \Rightarrow |P \cap \{1, \dots, n\}| = \left\lfloor \frac{n+1}{p} \right\rfloor.$$

$$a = \frac{n+1}{p} \notin \mathbb{Z}, \quad \frac{n+1}{q} = n+1 - a \Rightarrow [a] + [n+1-a] = n.$$

$$|(P \cup Q) \cap \{1, \dots, n\}| = n.$$

3

$$\forall n \geq 1, \quad (P \cup Q) \cap \{1, \dots, n\} = \{1, \dots, n\} \Rightarrow \mathbb{N}^* = P \sqcup Q.$$